

v2 Confirmation Run: Results

Reproducibility of survey question classification and harmonization findings across model generations

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Numbers source: every value in this document is transcribed from `v2/report/v2_report_numbers_collected.json` which records the WORK output file each number came from. Nothing here is computed in prose; the three derived percentages are computed in that JSON with their formulas recorded.

What this run is and why it exists

The v1 concept mapper classified roughly 7,000 questions from 47 Census Bureau demographic surveys into a shared topic and subtopic taxonomy, then coded harmonization barriers for candidate question pairs between source surveys (CPS, FoodAPS) and the ACS. Those findings were produced by one generation of models. If they cannot be reproduced by a different generation, they are an artifact of a particular model rather than a property of the surveys.

The v2 confirmation run reruns the pipeline over the same 6,987 questions with a new ensemble: `claude_4_5_sonnet` and `gemini-2.5-flash` as independent raters, arbitrated by `claude_4_5_opus`, all routed through the USAi harness. AHS was added as a net-new survey with no v1 baseline. Prompt equivalence between generations is gated by the TEVV prompt-equivalence check: 21 divergences between v1 and v2 prompts, every one acknowledged and justified, gate exit 0. Directional-bias hypotheses attached to those divergences are *provisional* pending SME review and are not findings of this report.

Headline findings

1. Topic-level classification reproduces across model generations. Final adjudicated agreement is 92.0 percent (kappa 0.880). Cross-generation raw-rater agreement (91.25 percent, best pair) exceeds v1's own within-generation rater agreement (89.24 percent), so the v1-to-v2 gap is no wider than the noise floor inside v1 itself.
2. Subtopic-level agreement is materially lower (76.9 percent final, kappa 0.761) and the confusion structure is the same in both generations: Social versus Economic at topic level, the Expenditures, Expenses, and Purchases cluster at subtopic level. The taxonomy boundary, not the model generation, is the binding constraint.
3. Disagreement concentrates where v1 was already uncertain. Questions v1 resolved by rater agreement reproduce at 97.5 percent topic agreement; questions from v1's arbitrated, new-concept, and unresolved strata fall to 85, 66, and 16 percent respectively.
4. The Stage 3 discard floor reproduces. Of v1's F3 discards on the common pair set, 96.7 percent (CPS) and 97.4 percent (FoodAPS) are discarded again by v2.

5. The candidate set partially turns over, in the strict direction. v2 retains 68 percent of v1 candidates; demotions to F3 outnumber promotions roughly four to one (46 versus 11 CPS, 27 versus 8 FoodAPS). The direction is consistent with the documented F3-attractor, which remains prompt-confounded and unresolved.
6. The dual-generation gold set is 62 candidate pairs (40 CPS, 22 FoodAPS): pairs both generations independently rated harmonizable at high confidence. This is the defensible bridge-variable lead list.
7. AHS, net-new in v2: of 736 unique AHS questions that entered pairing against ACS, 89 (12.1 percent) have at least one candidate match, 47 best-tier F1 and 42 best-tier F2. 259 of the 736 share no subtopic with ACS at all and are genuinely AHS-specific territory. Question counts are unique question texts; an id-based version of these figures (92 of 744) was retired as inflated. The AHS candidates carry single-generation provenance and are unvalidated leads, not results.

Stage 1: independent rater agreement (n = 6,987 questions)

The first question is whether individual raters from different generations even see the questions the same way, before any adjudication smooths the answer. The table compares every rater pairing: within v2, within v1, and across generations.

comparison	topic %	topic kappa	subtopic %	subtopic kappa
v2a vs v2b (within v2)	90.64	0.859	71.38	0.704
v1claude vs v1openai (within v1)	89.24	0.839	69.69	0.687
v1claude vs v2a (cross)	91.25	0.868	76.23	0.753
v1claude vs v2b (cross)	89.55	0.844	67.74	0.667
v1openai vs v2a (cross)	89.72	0.845	71.52	0.706
v1openai vs v2b (cross)	89.83	0.848	71.18	0.703

The cross-generation rows sit inside the same band as the within-generation rows. A v2 rater agrees with a v1 rater about as often as two v1 raters agree with each other, which is the operational definition of interchangeable raters. The subtopic ceiling near 70 to 76 percent appears in every pairing regardless of generation, which locates the limit in the taxonomy’s subtopic boundaries rather than in any model.

Stage 2: final adjudicated classification, v1 vs v2 (n = 6,987)

After arbitration resolves rater disagreements, each generation produces one final classification per question. This compares those finals head to head.

metric	value
topic agreement	92.03% (kappa 0.880)
subtopic agreement	76.90% (kappa 0.761)
full (topic + subtopic) agreement	76.00%
v1 unresolvable	0
v2 unresolvable	10

Reproduction stratified by how v1 reached its answer:

v1 decision method	n	topic agreement %
agreement	4,765	97.5
auto_dual_modal	821	85.1
pick_gpt5mini	522	84.7
pick_haiku45	482	83.4
new_concept	340	66.5
dual_modal	19	52.6
unresolved_disagreement	31	16.1
categorization_failed	7	14.3

The gradient is monotone in v1’s own certainty: the easier a question was for v1’s raters, the more faithfully v2 reproduces it. Classification instability is not random noise added by the new generation; it maps the questions the taxonomy never cleanly held. For the 68 percent of the corpus that v1 resolved by simple rater agreement, the v1 classifications can be treated as generation-stable.

Stage 3: harmonization barrier reproduction (CPS, FoodAPS vs ACS)

Pair universe intersection

Stage 3 operates on question pairs, and which pairs exist depends on the upstream subtopic assignments. Because subtopics shifted on roughly a quarter of questions, the v1 and v2 pair sets themselves only partially intersect. Every agreement number below is conditioned on the common pairs and is silent about the rest.

survey	v1 pairs	v2 pairs	common	Jaccard
CPS	603	659	480	0.614
FoodAPS	459	667	390	0.530

Agreement on common pairs

survey	n	L1 barrier % (kappa)	feasibility % (kappa)	both %
CPS	480	84.8 (0.518)	83.5 (0.604)	74.6
FoodAPS	390	92.6 (0.674)	88.5 (0.652)	83.6
combined	870	88.3 (0.576)	85.8 (0.625)	78.6

Raw agreement and kappa diverge because the class distribution is dominated by the CC barrier and the F3 tier, which inflates agreement-by-chance. Both numbers are reported and neither alone is the result.

Candidate dynamics on common pairs

The deliverable unit of Stage 3 is the candidate set: pairs rated F1 (direct recode) or F2 (statistical adjustment). This table tracks what happened to v1’s candidates under v2.

metric	CPS	FoodAPS
v1 candidates (F1/F2)	144	86
v2 candidates	109	67
intersection	98	59
demoted (v1 candidate, v2 F3)	46	27
added (v2 candidate, v1 F3)	11	8
binary candidate agreement	88.1% (kappa 0.696)	91.0% (kappa 0.717)
both-discard reproduction	96.7%	97.4%
gold set (dual-generation, high confidence)	40	22

Two asymmetries matter. First, discards are far more stable than candidates: when v1 said a pair was not harmonizable, v2 almost always agreed, so the negative finding is the most reproducible result in the study. Second, v2 is systematically the stricter rater, removing about a third of v1's candidates while adding few of its own. Whether that strictness reflects better judgment or an artifact of the acknowledged prompt divergences is exactly the question the TEVV directional-bias review (open, SME pending) exists to answer; this report does not prejudge it. The practical consequence either way: the 62-pair gold set, which requires agreement from both generations, is robust to the question.

AHS: net-new coverage (no v1 baseline)

metric	value
unique AHS questions entering pairing vs ACS	736
questions with at least one candidate (F1/F2)	89 (12.1%)
best tier F1 / F2	47 / 42
entered pairing, all pairs F3	388
no shared subtopic with ACS	259
candidate pairs / total pairs	130 / 1,626 (8.0%)
F3 pair share	92.0%

Question counts are unique question texts, the project's certified unit; 4 texts carrying duplicate ids previously inflated these figures to 92 of 744, and the id-based numbers are retired to a diagnostics block.

Reach is heavily structured by topic. Demographic questions reach 64.3 percent because AHS asks the same age, race, sex, and relationship questions ACS does; Housing, the bulk of the survey at 360 questions producing pairs, reaches 11.1 percent because AHS's housing content goes far deeper than ACS's. The 259 no-shared-subtopic questions are the cleanest finding: genuinely AHS-specific content with no ACS counterpart territory, which is the enrichment case for AHS in one number. Full subtopic tables, candidate listings, and the unvalidated-floor caveat are in `output/stage3/results/ahs/ahs_rollup.md` (WORK).

Limitations

The Stage 3 agreement numbers are conditioned on the common-pair intersection; roughly 40 percent of each generation’s pair universe has no counterpart in the other and is not assessed. AHS candidates are single-generation and unvalidated. The kappa values for Stage 3 reflect heavy class imbalance and should be read alongside raw agreement, not instead of it. The v2-stricter pattern is confounded with acknowledged prompt divergences until the TEVV directional-bias review closes. Ten questions v2 marked unresolvable had v1 classifications; they are counted as disagreements, not dropped.

Bottom line

The v1 findings survive the generation change where it matters. Topic classifications are generation-stable, discards reproduce at 97 percent, and the candidate signal concentrates into a 62-pair dual-generation gold set that no single model produced alone. The instability that does exist is informative rather than damaging: it sits on taxonomy boundaries and on questions v1 itself could not resolve, and v2’s stricter candidate judgment narrows the lead list rather than contradicting it.